SWE2106 Internet Technologies and Web Design

Etivity 5: Front end and Backend Development leveraging JavaScript

Instructions:

Due date: Wednesday 12th November 2026

Submission: Team activity presentation

Tools to be utilized: You have a variety of tools you can pick Pencil, LucidChart, Figma, or Draw.io), Text Editor (Sublime text, VS Code), and GitHub.

Preamble

The Academic Registrar at Mbarara University of Science and Technology has discussed digitizing the Academic Registrar department's work, with strong support from the Deputy Vice Chancellor. Your class has been contacted to provide proposals and solutions. While working in teams, your team will identify one registrar-related challenge and propose a practical digital solution.

Task

Choose a process such as examination results, application, student records, class attendance, transcripts, course registration, graduation clearance, or any registrar process. Discuss the problem as a team and propose a production-ready project: define its users, key features, and expected benefits. Build an Express server and API for the core workflow. Create a ***Postman*** collection to test successful and invalid requests.

Deliverables and Key timelines

Proposal: problem, users, objectives, scope, and solution - 10th November 2026.

1. Express server and API, with setup
2. Documented ***Postman*** collection with testable requests and sample data 12th November 2026
3. Team presentation with a live ***Postman API*** demonstration and project rationale - 12th November 2026.
4. Connect your project to a live ***Database*** and frontend
5. Github repository - 12th November 2026

NB: Demonstrate teamwork (with each key person's responsibility, GitHub pull request by each team member to show that they contributed), clear API design, input validation, and useful responses. Every team member should be ready to explain the design.